

AI/ML Internship – Day 49 Notes & Tasks

Module 6: Understanding Epochs, Training Results, Loss Graphs, Precision, Recall & mAP in YOLO Training

Part 1: Recap of Day 48

Yesterday we learned:

-  Custom Object Detection
-  Dataset Creation
-  Annotation
-  data.yaml
-  Training YOLO on Custom Dataset

Today we will learn how to analyze training results and determine whether our model is performing well.

Part 2: Why Evaluate a YOLO Model?

After training YOLO, we need to answer:

-  Is the model learning properly?
-  Is the model improving?
-  Is the model detecting objects accurately?

To answer these questions, we use:

- Epochs
 - Loss
 - Precision
 - Recall
 - mAP
-

Part 3: What is an Epoch?

Definition

An Epoch means one complete pass through the entire training dataset.

Example

Suppose:

100 Images

If YOLO sees all 100 images once:

Epoch = 1

If YOLO sees all images again:

Epoch = 2

Visual Representation

100 Images



Seen Once



Epoch 1

Seen Again



Epoch 2

Seen Again



Epoch 3

Part 4: Why Multiple Epochs?

One pass is usually not enough.

The model improves gradually.

Example:

Epoch 1 → Poor Detection

Epoch 10 → Better Detection

Epoch 50 → Good Detection

Part 5: What is Loss?

Definition

Loss measures how wrong the model's predictions are.

Simple Explanation

Higher Loss:

 More mistakes

Lower Loss:

 Better learning

Example

Epoch 1 → Loss = 5.0

Epoch 10 → Loss = 2.5

Epoch 20 → Loss = 0.8

This means:

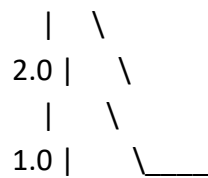
YOLO is improving.

Part 6: Loss Graph

Ideal Loss Curve:

Loss

```
5.0 | \
    | \
4.0 | \
    | \
3.0 | \
```



Epochs →

Important Rule

👉 Loss should decrease as training progresses.

📌 Part 7: Precision

Definition

Precision measures:

How many detected objects are actually correct.

Formula

$$\text{Precision} = \frac{\text{Correct Detections}}{\text{All Detections}}$$

Example

YOLO detects:

10 Helmets

Actual Helmets:

8 Correct

2 Wrong

Precision:

$$8 / 10 = 80\%$$

📌 Part 8: Recall

Definition

Recall measures:

How many actual objects were successfully detected.

Formula

Recall =
Detected Objects

Actual Objects

Example

Actual Helmets:

10

Detected:

8

Recall:

$8 / 10 = 80\%$

Part 9: Precision vs Recall

Precision

Accuracy of detections

Avoids false positives

Recall

Coverage of detections

Avoids missed objects

Easy Example

Precision

Detected = 10

Correct = 9

Precision:

90%

Recall

Actual Objects = 20

Detected = 15

Recall:

75%

Part 10: What is mAP?

Full Form

mAP = Mean Average Precision

Definition

mAP is the most important metric in Object Detection.

It combines:

 Precision

 Recall

into a single score.

Part 11: Why mAP is Important?

When comparing YOLO models:

Model A → mAP = 70%

Model B → mAP = 92%

Model B is better.

Part 12: Understanding mAP Values

mAP Performance

Below 50% Poor

50–70% Average

70–85% Good

Above 85% Excellent

Part 13: YOLO Training Output

During training you may see:

Epoch 1/50

Box Loss: 2.5

Cls Loss: 1.8

Precision: 0.70

Recall: 0.65

mAP50: 0.72

Part 14: Understanding These Metrics

Box Loss

Bounding box accuracy.

Lower is better.

Cls Loss

Classification error.

Lower is better.

Precision

Higher is better.

Recall

Higher is better.

mAP

Higher is better.

Part 15: Example Training Progress

Epoch 1

Loss = 5.0

Precision = 40%

Recall = 35%

mAP = 30%

Epoch 20

Loss = 1.5

Precision = 80%

Recall = 78%

mAP = 82%

This means:

Model is learning successfully.

Part 16: Training Result Files

After training:

runs/

└─ detect/

train/

Inside:

results.png

best.pt

last.pt

Part 17: What is best.pt?

Best-performing model.

Use this for deployment.

Part 18: What is last.pt?

Model from final epoch.

Not always the best.

Part 19: Understanding results.png

YOLO automatically creates graphs.

Example:

Loss Curve

Precision Curve

Recall Curve

mAP Curve

Ideal Result

Loss ↓

Precision ↑

Recall ↑

mAP ↑

Part 20: Real-World Example

Helmet Detection Project

Training Result:

Precision = 95%

Recall = 92%

mAP = 94%

Meaning:

Excellent model.

Suitable for deployment.

Part 21: Common Mistakes

Too Few Epochs

Epochs = 2

Model may not learn properly.

Very Small Dataset

Poor accuracy.

Bad Annotation

Incorrect bounding boxes.

Poor performance.

Overtraining

Too many epochs can cause:

Overfitting

Part 22: Theory Questions

What is an Epoch?

One complete pass through the dataset.

What is Loss?

Measure of prediction error.

What is Precision?

Correct detections among all detections.

What is Recall?

Detected objects among actual objects.

What is mAP?

Mean Average Precision.

Main evaluation metric in object detection.

Tasks for Day 49

Theory

1. What is an Epoch?
2. What is Loss?

3. What is Precision?
 4. What is Recall?
 5. What is mAP?
-

Practical Task 1

Train a YOLO model.

Observe:

Epoch

Loss

Precision

Recall

mAP

Practical Task 2

Locate:

best.pt

last.pt

results.png

Practical Task 3

Analyze training graphs.

Practical Task 4

Compare:

Epoch 1

vs

Epoch 10

Practical Task 5

Explain why loss decreases during training.